# Module0-Question2

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## Definition of Variable

- Response Variable (prog.comf): Students' self-rated programming comfort level (measured on a numeric scale, typically from 1 to 5).
- Factor 1 (research): Prior research experience, categorized as Yes, No, or Unsure.
- Factor 2 (domain_specialization): Domain specialization status, categorized as Yes, No, or Unsure.

## Preparation for data

```
library(tidyverse)
library(ggplot2)
library(dplyr)

# read data
background <- read_csv("../data/background-clean.csv")
interest <- read_csv("../data/interest-clean.csv")

# joins
df_merged <- inner_join(background, interest, by = "response_id")


# Prepare data for analysis
anova_data <- df_merged %>%
  select(response_id, prog.comf, research, domain_specialization) %>%
  drop_na()
```

## Summary statistics for each combination

```
group_summary <- anova_data %>%
  group_by(research, domain_specialization) %>%
  summarise(
    n = n(),
    mean_comfort = mean(prog.comf),
    sd_comfort = sd(prog.comf),
    .groups = "drop"
  )

group_summary
```

```
## # A tibble: 8 x 5
##   research domain_specialization     n mean_comfort sd_comfort
##   <chr>    <chr>                 <int>        <dbl>      <dbl>
## 1 No       No                       12         3.67      0.778
## 2 No       Unsure                    3         3         1
## 3 No       Yes                       7         3.57      0.535
## 4 Unsure   No                        3         3         0
## 5 Unsure   Yes                       1         3        NA
## 6 Yes      No                        9         3.78      0.441
## 7 Yes      Unsure                    4         3.75      0.5
## 8 Yes      Yes                      10         4.2       0.919
```

```
# Check all combinations, including empty groups
with(anova_data, table(research, domain_specialization))
```

```
##          domain_specialization
## research No Unsure Yes
##   No     12      3   7
##   Unsure  3      0   1
##   Yes     9      4  10
```

**Interpretation**:

The analysis includes 49 students across eight observed combinations of research experience and domain specialization. Group sizes are unequal, ranging from 1 to 12 students, and no students selected "Unsure" for both factors. Students who reported both research experience and domain specialization had the highest mean programming comfort score (4.20).

Within each domain specialization category, students with research experience reported higher average programming comfort than those without research experience. However, these differences are descriptive; statistical testing is needed to assess their significance. The small sample sizes in several groups also require cautious interpretation.

## Two-way ANOVA

```
anova_model <- aov(
  prog.comf ~ research * domain_specialization,
  data = anova_data
)

summary(anova_model)
```

```
##                                Df Sum Sq Mean Sq F value Pr(>F)
## research                        2  3.997  1.9985   4.039 0.0251 *
## domain_specialization           2  1.252  0.6261   1.265 0.2929
## research:domain_specialization  3  0.872  0.2908   0.588 0.6266
## Residuals                      41 20.287  0.4948
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Interpretation**:

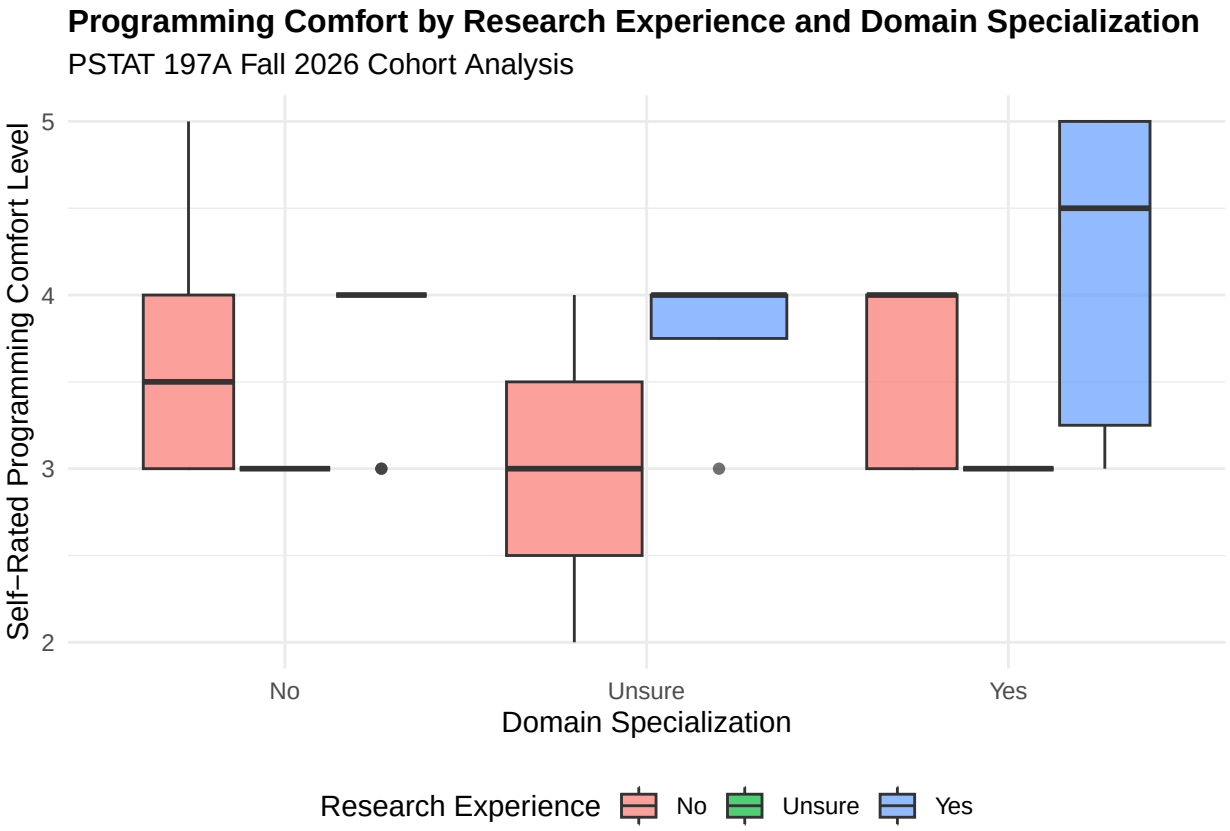
The two-way ANOVA showed a statistically significant result for research experience in the sequential analysis, $F(2, 41) = 4.039$, $p = 0.0251$. However, this overall test does not identify which research experience groups differ. Domain specialization was not statistically significant when entered after research experience, $F(2, 41) = 1.265$, $p = 0.2929$.

The interaction between research experience and domain specialization was also not statistically significant, $F(3, 41) = 0.588$, $p = 0.6266$. Thus, there is insufficient evidence that differences in programming comfort across research experience groups vary by domain specialization. This does not establish that no interaction exists, particularly given the small sample sizes in several groups.

Because the group sizes are unequal, the sequential main-effect tests may depend on the order of the factors. The research experience result, which was tested first, should not be interpreted as an effect adjusted for domain specialization.

## Plot

```
ggplot(anova_data, aes(x = domain_specialization, y = prog.comf, fill = research)) +
  geom_boxplot(alpha = 0.7, position = position_dodge(0.8)) +
  labs(
    title = "Programming Comfort by Research Experience and Domain Specialization",
    subtitle = "PSTAT 197A Fall 2026 Cohort Analysis",
    x = "Domain Specialization",
    y = "Self-Rated Programming Comfort Level",
    fill = "Research Experience"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 12),
    legend.position = "bottom"
  )
```




**Interpretation**:

The boxplot shows that students with research experience have higher median programming comfort than those without research experience within each domain specialization category. Students reporting both research experience and domain specialization have the highest median comfort score, approximately 4.5, although their scores also show considerable variation.

The distributions overlap across several groups. Some boxes collapse to a single line because of limited variation in scores or very small group sizes. No box appears for students who selected "Unsure" for both factors because this combination has no observations.

These visual patterns suggest an association between research experience and programming comfort, but the plot alone does not establishstatistically significant differences or an interaction.